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1. Explain the role of Tokenization in Natural Language Processing.

Ans:- The role of Tokenization in Natural Language Processing (NLP)

- Breaks text into smaller units like words, subwords, or characters.
- Prepares raw texts for further NLP processing.
- Helps in vocabulary creation for language models.
- Enables text representation in numerical form.
- Improves model understanding of sentence structure.
- Handles punctuation and special symbols effectively.
- Supports different languages with varied writing systems.
- Reduces complexity in text analysis tasks.
- Essential for tasks like translation, sentiment analysis, and chatbots.
- Affects model accuracy and performance directly.

2. Analyze the differences between syntax, semantics, and pragmatics in the context of natural language understanding.

Ans:- 1. Syntax (Structure of language)

- Deals with the grammatical structure of sentences.
- Focuses on word order and sentence formation rules.

- Does not consider meaning, only correctness of structure.

2. Semantics (Meaning of language)

- Deals with the literal meaning of words and sentences
- Interprets what a sentence actually means
- Ensures meaningful understanding beyond grammar

3. Pragmatics (Context of language)

- Deals with the intended meaning based on context
- Considers speaker intention, situation, and real-world knowledge
- Helps understanding implied or indirect meanings

3) Illustrate the application of K-Nearest Neighbour (K-NN) algorithm in document classification:

Ans:- 1. Introduction to KNN:- K-Nearest Neighbour (K-NN) is a supervised machine learning algorithm, used for classification tasks. It classifies a new data point based on majority class.

2. Purpose in Document Classification:- K-NN is used to assign category labels to new text documents based on similarity with already labeled documents.

3. Document Representation:-

Before applying K-NN, documents are converted into numerical features vectors using techniques such as:-

- Bag of Words (BoW)
- TF-IDF (Term frequency - Inverse Document frequency)

4. Measuring Similarity:-

- Cosine similarity (most common for text)
- Euclidean distance

5. Selection of K.

- Small K - Sensitive to noise
- Large K - More stable but computationally heavier.

6. Classification Process:-

1. Convert the new document into a vector.
2. Compute similarity with all training documents.
3. Select the K closest documents.
4. Assign the class most frequent among these neighbours